Work Integrated Learning Programmes Division

M.Tech. in AIML

Natural Language Processing

S2-25_AIMLCZG530

Assignment 1

Problem Statement - 13

General Instructions

1. **Implementation on the BITS OSHA Virtual Lab is compulsory, and carries 1 mark.**
2. **Submit the code in .pdf/html format along with output. Please follow the naming convention as <Group no>.pdf. Example for group 1 should be named as - Group1.pdf.**
3. **Include group details like team members' names, BITS ID, and contribution percentages.**
4. **All members of the group will work on the same problem statement. Each group should upload in Taxila in respective locations under ASSIGNMENT Tab. Assignments submitted via means other than through Taxila will not be graded.**
5. **One student per group is requested to submit your assignment.**
6. **Any queries related to this problem statement should be addressed to Vasugi I (vasugii@wilp.bits-pilani.ac.in), Course LF.**
7. **If the dataset link is not working, download a similar dataset from an online resource and note this in your submission.**

LSTM Language Model for Text Generation

Link to the Dataset: https://www.kaggle.com/datasets/shivamb/shakespeare-sonnets

Description of Data: Shakespeare Sonnets dataset with 5,028 lines of poetry. Suitable for character-level and word-level text generation.

1. Download the file and set it as a DataFrame. Combine all text into a single string. (1 Mark)

2. Create character-level vocabulary. Encode text as integer sequence. Create sequences of length 50 for training. (2 Marks)
3. Implement a 2-layer LSTM model (hidden size = 256). Display LSTM architecture and number of trainable parameters. (3 Marks)
4. Train model for 20 epochs. Generate 100 characters of text given seed string "Once upon" and display the generated Shakespeare-style text. (3 Marks)
5. Compare and contrast RNN/LSTM language models vs N-Gram models for text generation. How does LSTM handle long-term dependencies that N-Gram models cannot capture? (2 Marks)